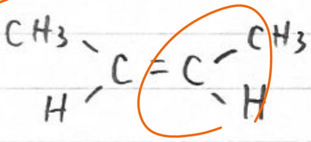
No.
Date . .

2-⑥ Ⅱ

1、ア 酸化銅；試料の不完全燃焼を防ぐ
　イ 塩化カルシウム；水を吸収する
　ウ ソーダ石灰；二酸化炭素を吸収する

2、C_4H_8

3、A 

B $CH_2=CH-CH_2-CH_3$

C $CH_2=C(CH_3)-CH_3$

D $CH_3-CH(OH)-CH_2-CH_3$

E $CH_3-C(OH)(CH_3)-CH_3$

F $CH_3-C(=O)-H$

G $H-C(=O)-H$

H $CH_3-CH_2-C(=O)-H$

I $CH_3-C(=O)-CH_3$

4、F

3-⑭

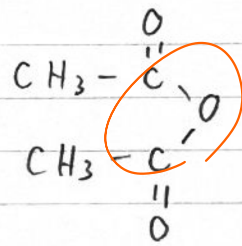
1、A $CH_3-CH_2-O-C(=O)-CH_3$

B $(CH_3)_2CH-O-C(=O)-H$

C CH_3-CH_2-OH

D $CH_3-C(=O)-OH$

E $CH_3-C(=O)-H$

F 

G $CH_3-C(=O)-CH_3$

H $H-C(=O)-OH$

I $CH_3-CH(OH)-CH_3$

2、ア 水酸化ナトリウム　イ ヨウ素
　ヨードホルム反応；CHI_3

3、銀鏡反応

4、アセトアルデヒド

3-③

1、(a) 8種類
　(b) 6種類

2、3種類

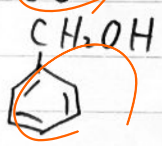
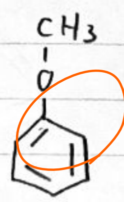
3、4種類

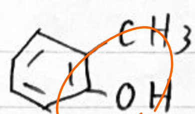
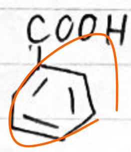
4、X $CH_3-CH_2-CH_2-CH_2-CH_2-C(=O)-H$

Y $CH_3-CH_2-CH_2-CH_2-CH_2-C(=O)-OH$

4-③

1、C_7H_8O

2、A  B 

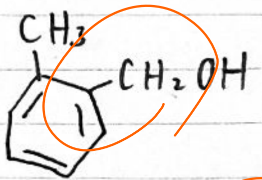
C  D 

No.

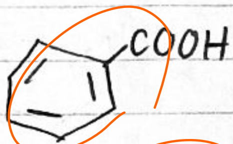
Date

4-$\boxed{14}$

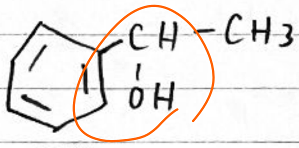
(1) ~~4~~ (2)



(3) CHI_3 (4)

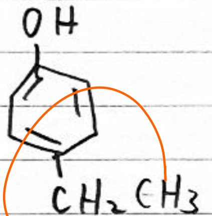


(5)



(6) 9

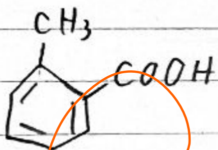
(7)



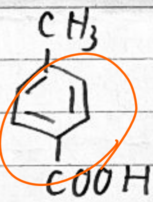
(8) 5 (9) 1:2:2:2:3

4-$\boxed{16}$

1、(1) A

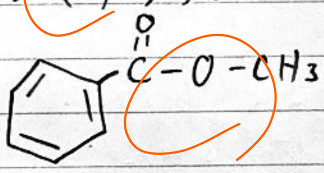


B

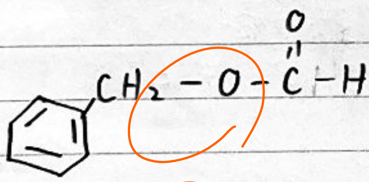


(2) o-キシレン、ナフタレン

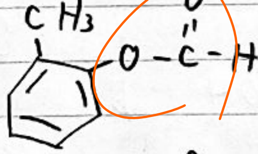
2、(1) C



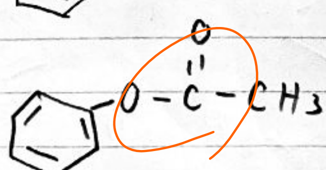
D



E



F

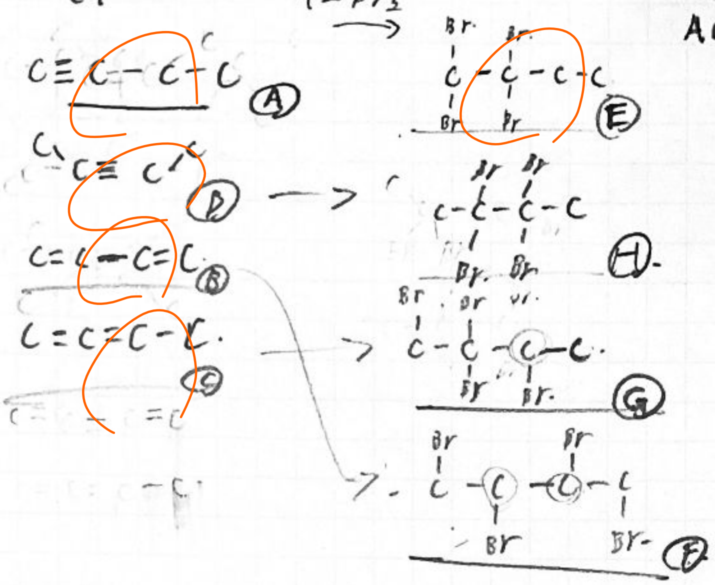


(2) ~~濃硫酸~~

2-$\boxed{14}$

(a) A, B, C, D, E

C_4H_6 $+2Br_2$



$\overset{+}{Ag}NO_3$.

ヨードホルム

Aの時 $\xrightarrow{\text{小さな}HgO}$ $H_3C-C(=O)-R$ or $H_3C-CH(OH)-R$

4-$\boxed{12}$ C_7H_8O

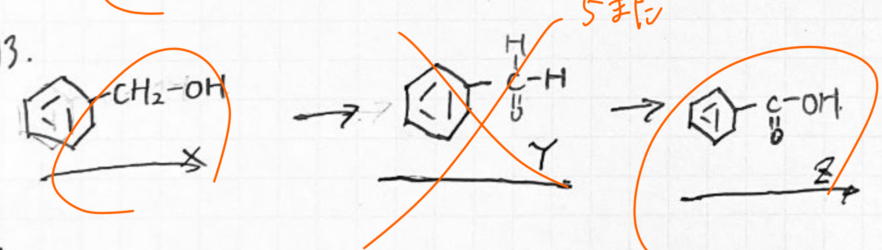
問1 操作1で発生した気体 H_2

→ 金属ナトリウムと反応 $-OH$ or ~~$-COOH$~~

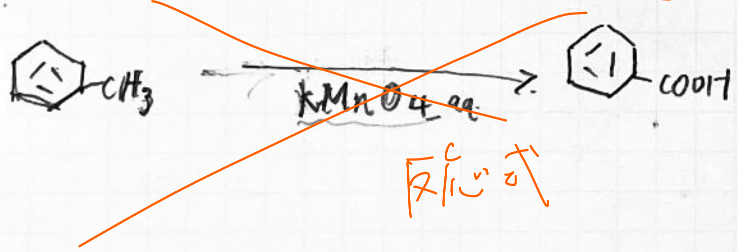
問2 フェノール類の検出 塩化鉄(III) $FeCl_3$

青色～青紫色

問3.



問4.



反応式

3-[15]

$C_4H_8O_2$

カルボン酸エステル　NaOHaq と加熱　ホルミル A D　ヨードー A C

$\xrightarrow{\text{けん化}}$

C-C-C-C-OH

C-C-C-OH

たぶん

C-C-C-O-C-H (D)　　C-C-C-ONa + HO-C-H

C-C(C)-O-C-H (A)　　C-C(C)-ONa + HO-C-H　A

C-C-O-C-C (C)　　C-C-ONa + HO-C-CH₃　C

C-O-C-C-C (B)　　C-ONa + HO-C-C-C

C-C-C-ONa + H₂O

C-C(C)-C-ONa + H₂O

4-[18]

(1) C　H　O.

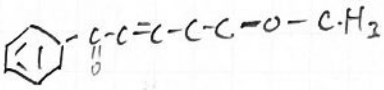
75.8%　7.4%　16.8%

$\frac{75.8}{12} : \frac{7.4}{1} : \frac{16.8}{16}$

6.15 : 7.4 : 1.05

14 : 15 : 2

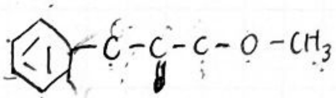
$C_{12}H_{14}O_2$

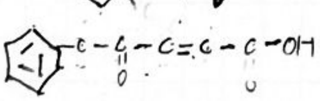


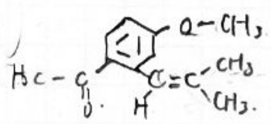
(2)

加水分解

CH_3-OH

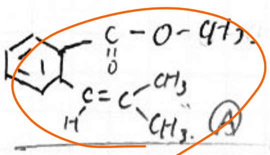


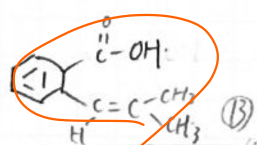




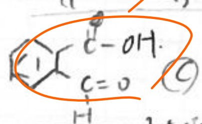
$U = \frac{12\times2+2-14}{2} = 6$

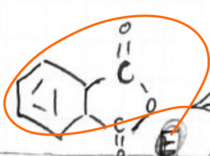
ベンゼン 4つ
他 2つ



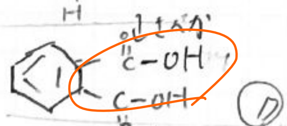


→ オゾン分解 →　$CH_3-C(=O)-CH_3$





脱水



2-$\boxed{2}$

(1) $\frac{4}{160} = 0.025$ mol　　$\frac{3.21}{214} = 0.15$ mol (CH_2)

$\frac{0.15}{n}$ mol の $(CH_2)_n$ に 0.025 mol の Br_2 が付いた

$\Rightarrow \frac{0.15}{n} = 0.025$　∴ $n = 6$

C_6H_{12}

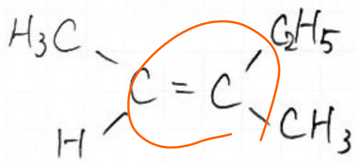
(2)

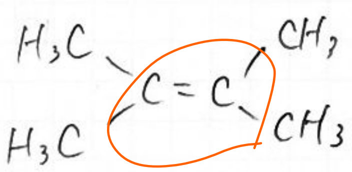
A $\xrightarrow{O_3}$ $CH_3-\underset{\overset{\|}{O}}{C}-H$ + $R-\underset{\overset{\|}{O}}{C}-R$

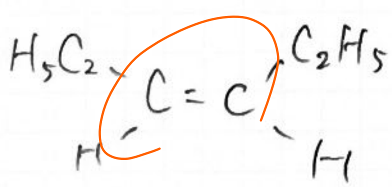
B $\xrightarrow{O_3}$ $R-\underset{\overset{\|}{O}}{C}-R$ ×2

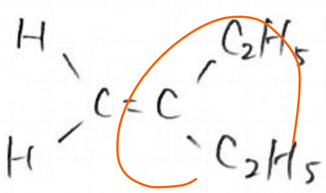
C $\xrightarrow{O_3}$ $R-\underset{\overset{\|}{O}}{C}-H$ ×2

D $\xrightarrow{O_3}$ $H-\underset{\overset{\|}{O}}{C}-H$　　対称的　$R-\underset{\overset{\|}{O}}{C}-R$

A :


B :


C :


D :


(3) A, C

3-$\boxed{7}$

A $\xrightarrow{NaOH}$ Bの Na塩 + C(中性)
(116)
Br_2と反応

B $\xrightarrow{\Delta}$ D + H_2O

C (アルコールX)(還X)(ヨO)

(1) C : H : O
$= \frac{53.8}{12} : \frac{5.1}{1} : \frac{41.1}{16}$
$= 4.5 : 5.1 : 2.5$
$= 9 : 10.2 : 5$

4.4 / 12)53.8 / 48 / 58　　2.5 / 16)41.1 / 32 / 9.1 / 80 / 11　　$\frac{16}{5}$　　12 / 9 / 108 / 90 / 198

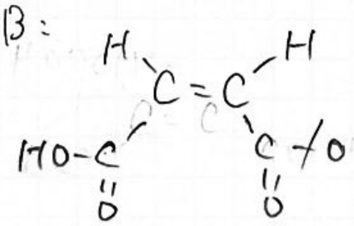
~~$(C_9H_{10}O_5)_n$~~ or $(C_2H_2O)_n$　$M = 42n$

24 / 16 / 2 / 42

$n = 4$

~~$C_8H_8O_4$~~　$M = 168$

(2) 不 $= \frac{16+2-8}{2} = 5$

B :

$CH_3-\underset{\overset{\|}{O}}{C}-CH_2CH_3$

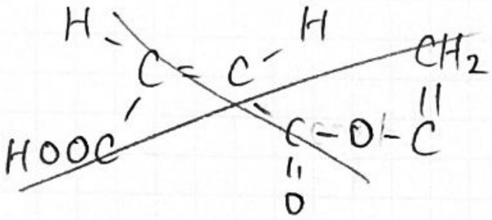
~~C-C-C-C~~

$48+64+4$
$= 116$

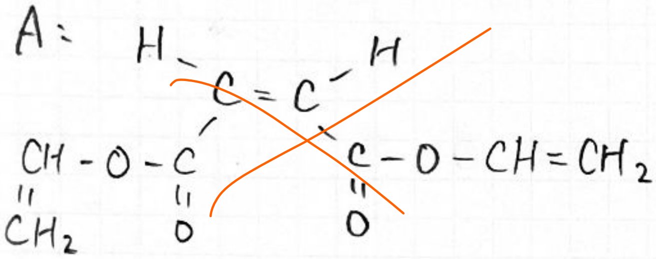
~~HO-C-C-C-C~~

HO-

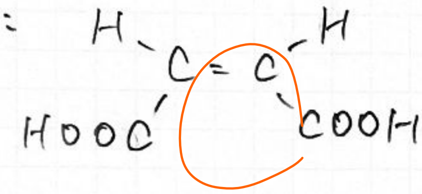
B:



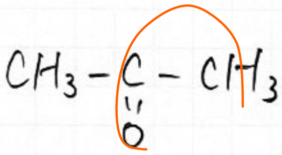
A:



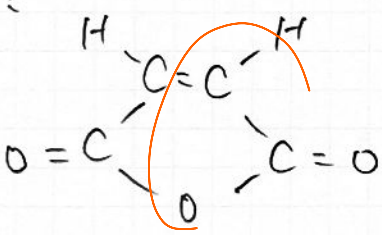
B:



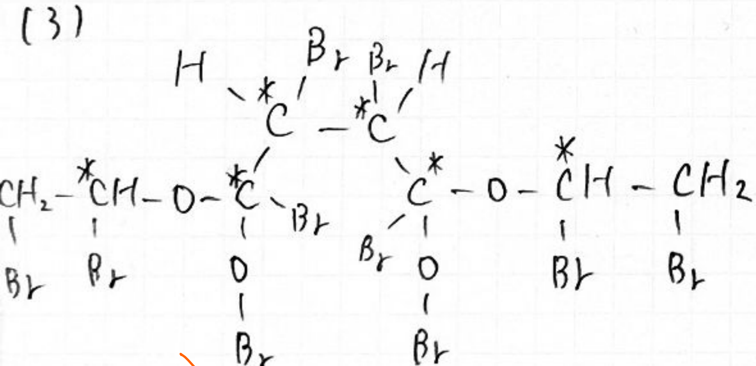
C:



D:



(3)



$2^6 = 64$ 〃

3-$\boxed{10}$

$C_4H_{10}O$　㊄ $= \dfrac{8+2-10}{2} = 0$

A（アルコール○）

B（ 〃 ）

C（ 〃 ）

D（ 〃 ）$\xrightarrow[\Delta]{H_2SO_4}$ F（2重結合あり）

E（アルコール×）

$C_2H_5OH \times 2 \rightarrow$ E $C_2H_5-O-C_2H_5$

(1)

A $CH_3CH_2CH_2CH_2OH$ 〃

B $CH_3CH_2CH(OH)CH_3$ 〃

D $CH_3-C(CH_3)(OH)-CH_3$ 〃

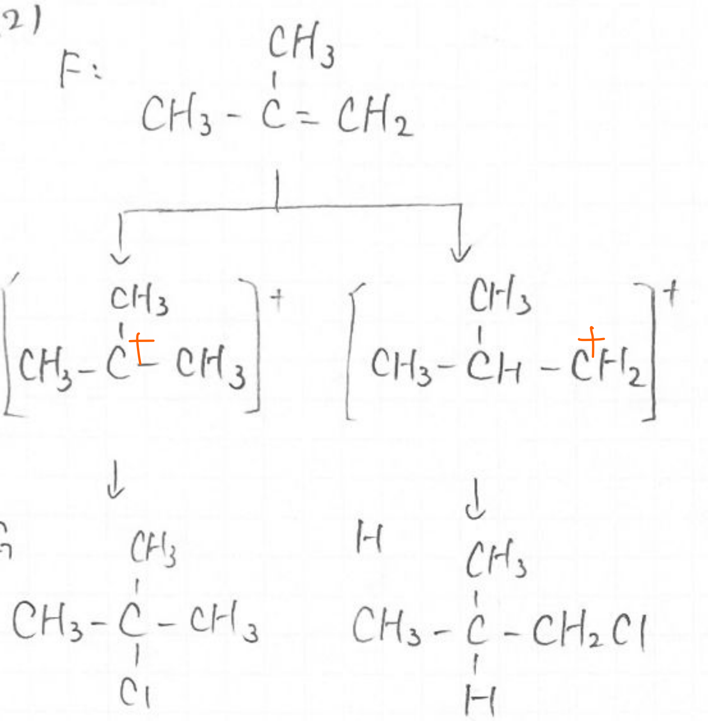
C $CH_3-O-CH_2CH_2CH_3$ 〃

E $C_2H_5-O-C_2H_5$ 〃

3-10

(2)

F:



(3)

$$\cancel{CH_3-CH(OH)-CH_2-CH_3 + 4NaOH + 3I_2}$$

$$\to CH_3-CH(ONa)-CH_2-CH_3 + 3H_2O$$

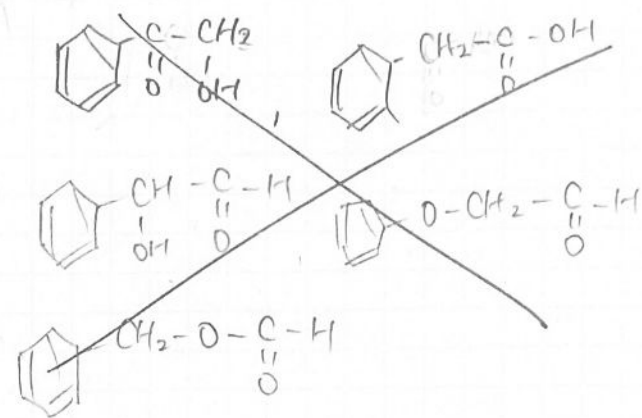
$$+ CHI_3 + 3NaI$$

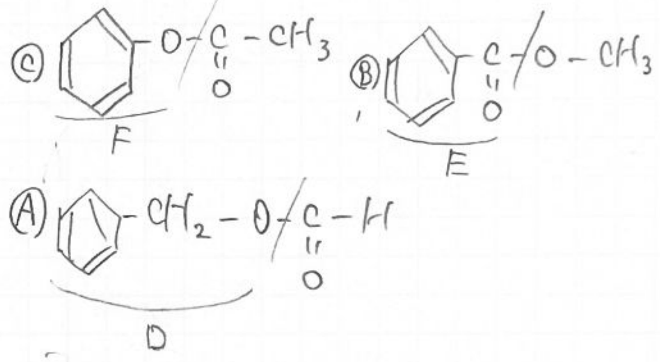
(4) ~~B~~

(5) エーテル結合

4-7

$C_8H_8O_2$ ㋐ $= \frac{16+2-8}{2} = 5$





(1)

A

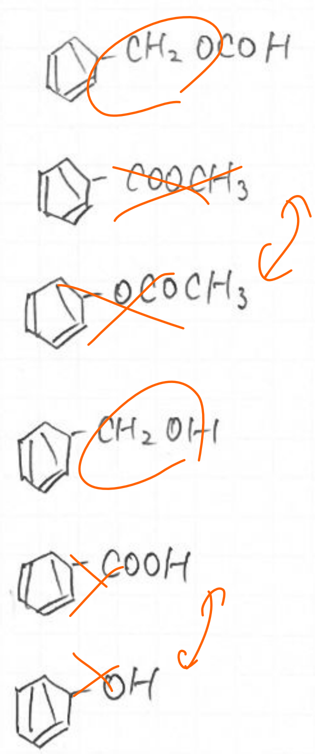
B

C

D

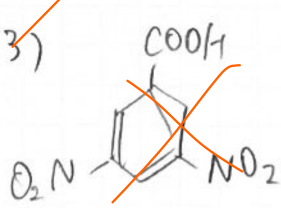
E

F



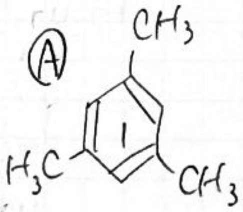
(2) ~~F~~

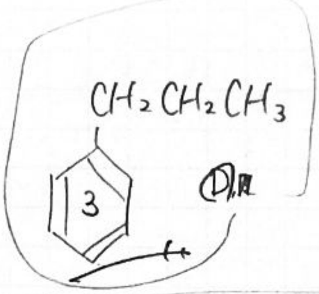
(3)



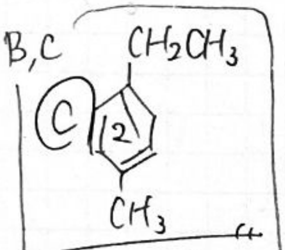
3,5-ジニトロ安息香酸

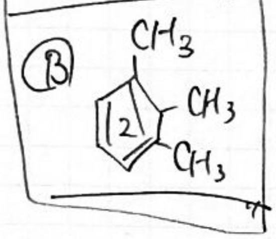
4-10



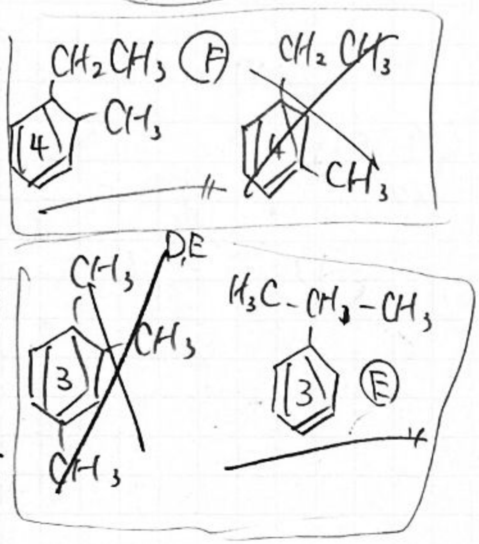


B,C





D,E



B $\xrightarrow{O_x}$ G

C $\xrightarrow{O_x}$ H

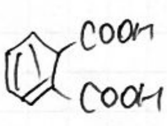
D $\xrightarrow{O_x}$ I

F $\xrightarrow{O_x}$ J

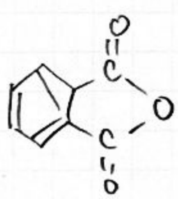
異置換体

同置換体

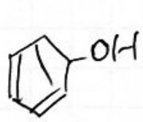
J



K



L



M $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$

(1)

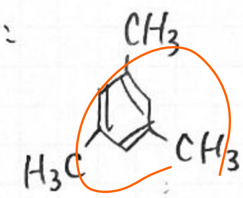
(ア): 1,2-エタンジオール

(イ): 酸素

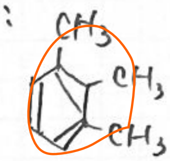
(ウ): ヨウ素

(2)

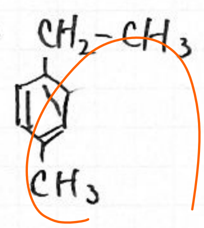
A:



B:



C:



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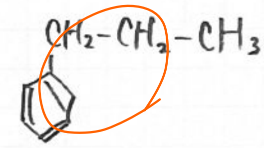

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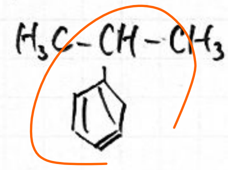

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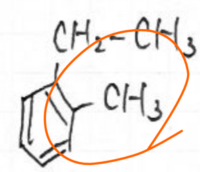
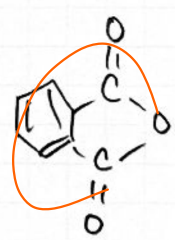


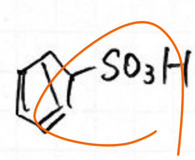
K:



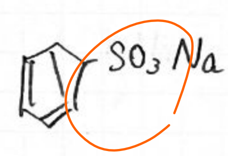
M: $H_3C-\underset{\underset{O}{\|}}{C}-CH_3$

(3)

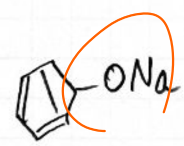
N:



O:



P:

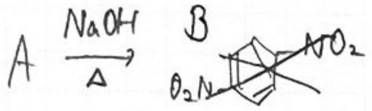


4-[20]

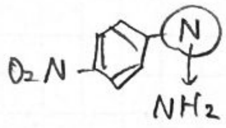
A:(カルボン酸)

㋐ $= \frac{22+2-12+2}{2} = 7$

$14+32$

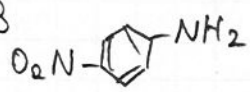


$\cancel{76+92=16}8$



$N = 138-76-46$
$= 16$

B



(1) $C:H:N:O$

$= 132:12:28:80$

$132 + 12 + 28 + 80 = 252$

C … 52.4%　　O … 31.7%

H … 4.8%

N … 11.1%

(2)

$RCOOH + NaHCO_3$

$\rightarrow RCOONa + H_2O + CO_2$

(3)

㋐ $7 = 4 + 1 + 2$
　　べ　カル

A

Ⓐ ＊1コ

$C_{11}H_{12}N_2O_5 + H_2O$

$\rightarrow$ Ⓑ $C_6H_6N_2O_2$ + Ⓒ ＊なし $C_5H_8O_4$

＊なし　　-COOH あり ×2

$U_c = \frac{10+2-8}{2} = 2$

$C_5H_8O_4$

$HOOC-CH_2-CH_2-CH_2-COOH$

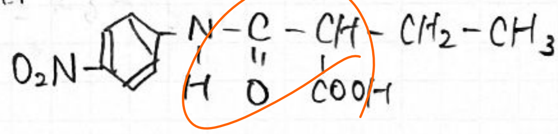
$\cancel{CH_3-{}^{*}CH(COOH)-CH_2}-COOH$

C

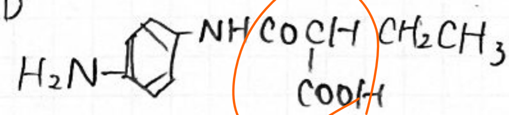
$\boxed{CH_3-CH_2-CH(COOH)-COOH}$

$CH_3-C(COOH)(COOH)-CH_3$

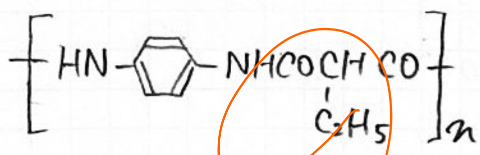
A



D



E



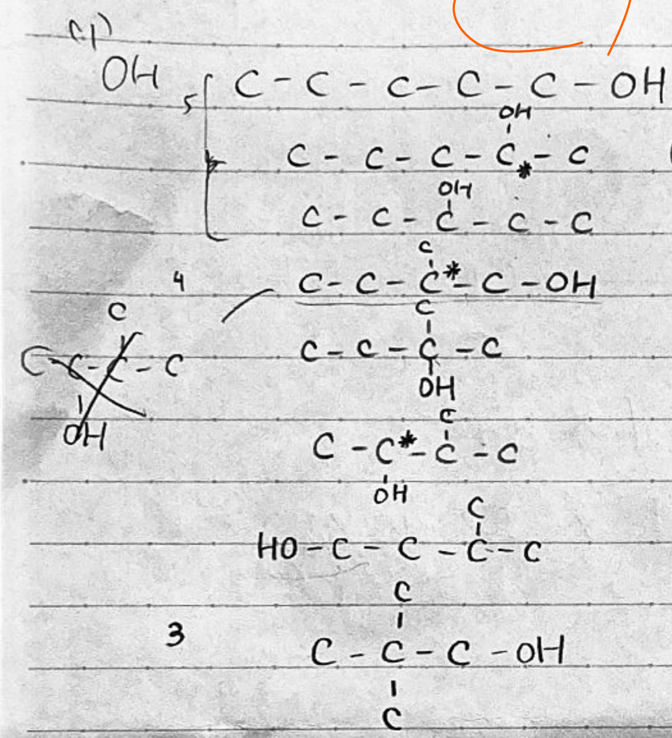
(4) アミド結合が塩酸により加水分解してしまうから。

No.

Date

3 - $\boxed{2}$ $C_5H_{12}O$

(1)



Q1 A

ヨード

Q 2 E, G

Q2 F

Q1 B

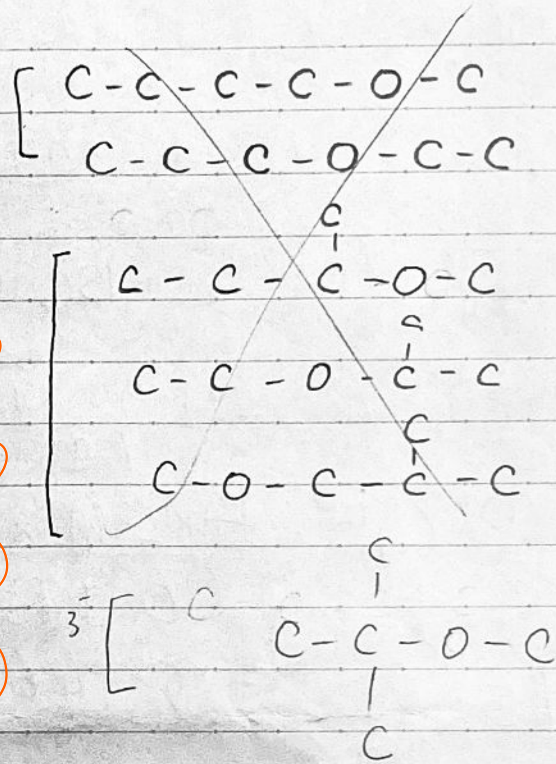
Q3 →H

Q2 E G

ヨード

Q1 C

Q1 D



(2) 6

○

3 - $\boxed{15}$

$C_5H_{10}O_2$ 分子量 60+10+32

96 × 13 = 288, 96 → 1248

106) 1020 → 0.962
959
660
636
240

反応式

(1) 酪酸メチル　$CH_3-CH_2-CH_2-C(=O)-O-CH_3$

硫酸：~~脱水~~の促進のため。
エステル化の触媒

(2) $2C_5H_{10}O_2 + 13O_2 \rightarrow \cancel{5}^{10}CO_2 + \cancel{5}^{10}H_2O$

酪酸メチル → $\dfrac{1.02 \times 10^{-1}}{102} = 1 \times 10^{-3}$ mol

O_2 (L)　$\dfrac{13}{2} \times 1 \times 10^{-3}$ mol × 22.4 L/mol

= 0.1456 L

≒ 0.146 L

224 × 13 = 672, 224 → 0.2912

(3) A　$C-C(=O)-O-C-C-C$

B　CH_3COOH

C　C_3H_7OH

↓ 酸化

D　$(CH_3CO)_2O$

E　$C-C-C(=O)-H$

(3) B 酢酸　C 1-プロパノール　A

$CH_3-C(=O)-O-CH_2-CH_2-CH_3$

(4)
$C-C(OH)-C$
$C-C-O-C$

(5) D　$CH_3-C(=O)$
$\quad O$
$CH_3-C(=O)$

(6) ホルミル基

Q

24

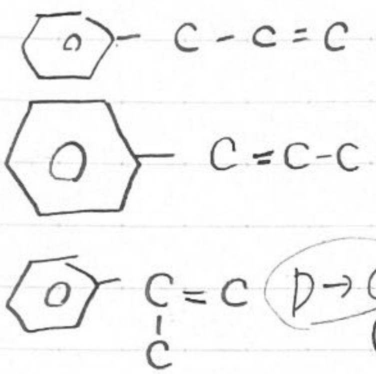
4-$\boxed{2}$

C_3H_4

C_9H_{10}

①

C_3H_5 (そくさ)



C

A/B

trans

D→G + H_2

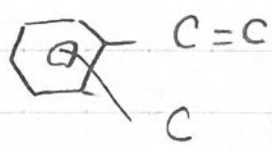
(F, G, H)

になる

数があわなかったら
異性体を考える。

②

C_3H_6



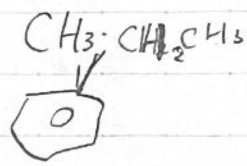
(3)

o m p

E
↓
H

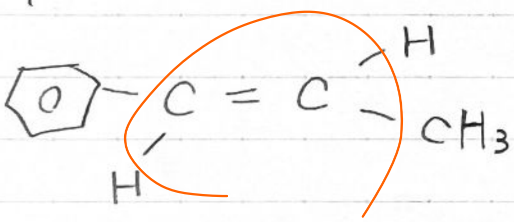
6

$CH_2=CH-CH$

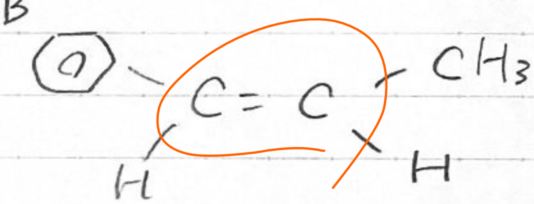


(1) ~~6~~ 8 種類

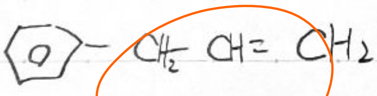
(2) A



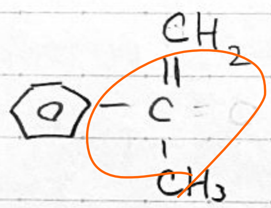
B



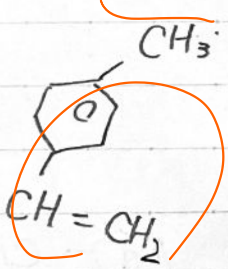
C

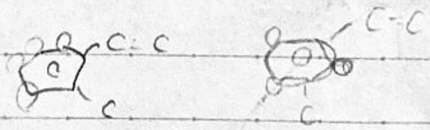


D



E



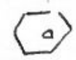


(3) 6 + 8 + 2 = 16

6 + 10 + 6 = 22

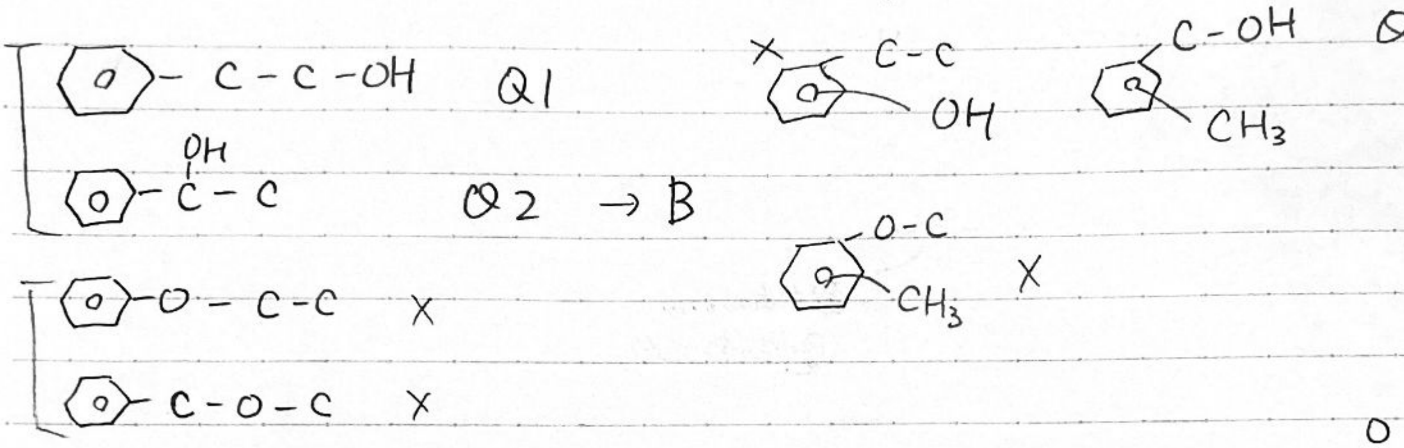
C

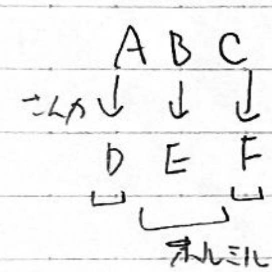
4-$\boxed{15}$

$C_8H_{10}O$  -4 C_2H_4O

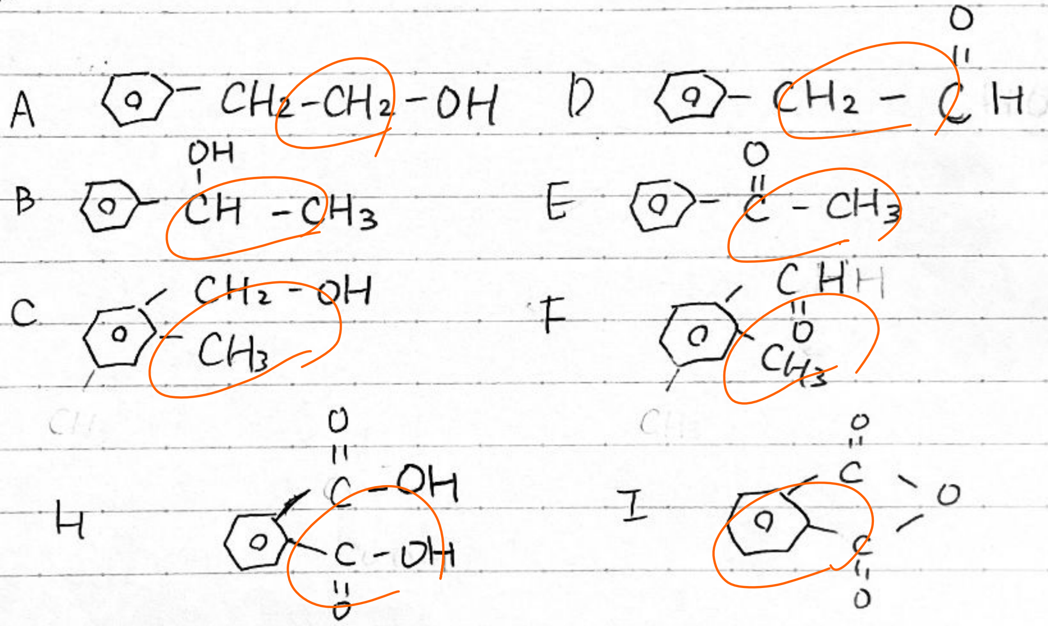
① C_2H_5O

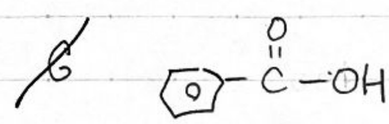
② C_2H_6O



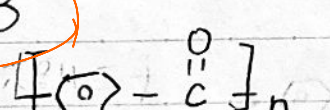


(1)



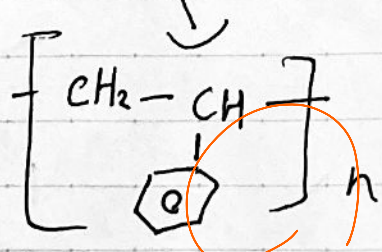


(2) B

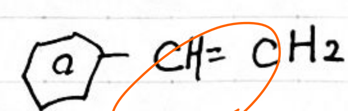
~~(3)~~ 

ポリフェニル酢酸

(4) B CHI_3



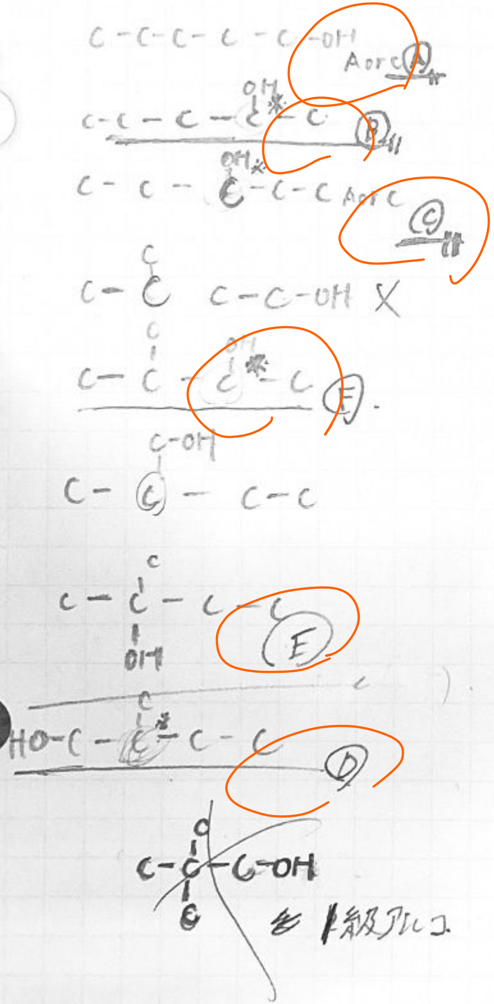
ポリスチレン



脱水 → 濃硫酸 + 加熱
$KMnO_4$ + 加熱
側鎖がカルボン酸に

まちがった理由

3-[12] $C_5H_{12}O$ オールアルコ　〇が不斉炭素　BDF



ABC 枝分かれなし。

DF ヨードホルム

Eのみ酸化しにくい。

ADの酸化したものと 銀NO_3aq が反応。

ホルミル基.

問2.

Ⓐでやる。 2C-C-C-C-C-OH + 2Na → 2C-C-C-C-C-ONa + H_2.　Ⓐ < Na

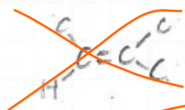
Na $\frac{0.3}{23}$ = 0.013… mol　$\frac{10}{88}$ = 0.125　0.013… mol 反応

$\frac{0.3}{23} \times \frac{1}{2} \times 22.4 = \cdots$ ≒ 0.15 L

問3. 銀鏡反応

ホルミル基　R-C(=O)-H

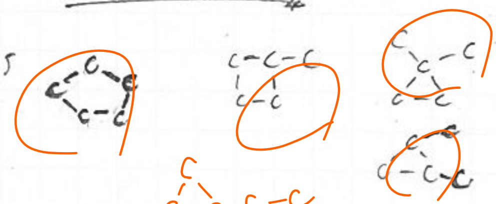
問4. C_5H_{10}



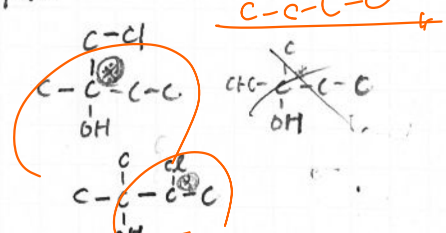
C=C-C-C-C

C-C=C-C-C

問5



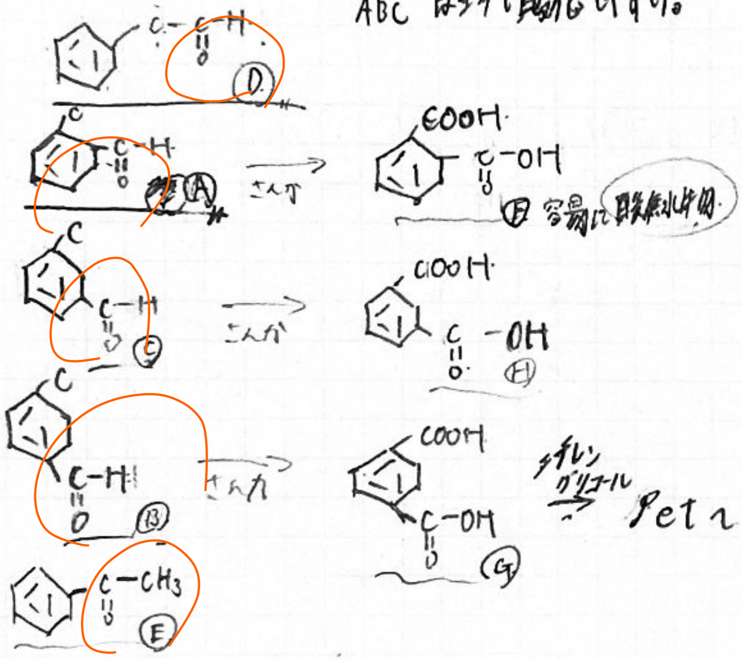
問6



1級アルコ

4−5. C_8H_8O カルボニル基 $-\underset{\underset{O}{\|}}{C}-OH$

ABC は容易で酸化しやすい。



F, H は官能基の配置的に分子内で水素結合が起きやすいが G はパラの配置であり他の分子と水素結合を起こしやすいから。

G は F, H より
対称性があり、
結晶化しやすい
→ 融点高い

4-1(1)

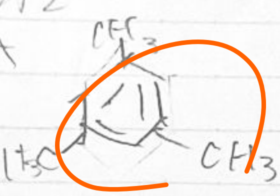
問1

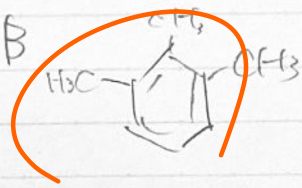
ア エチレン グリコール

イ 酸素

ウ ヨウ素

問2

A


B


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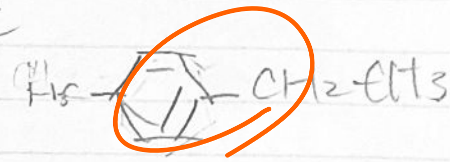
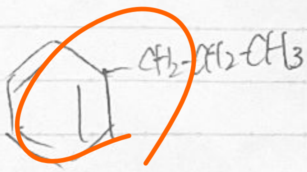


D


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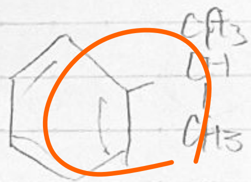
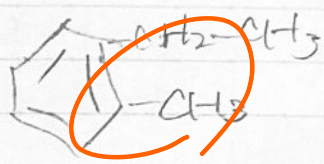
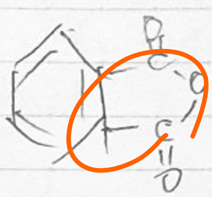
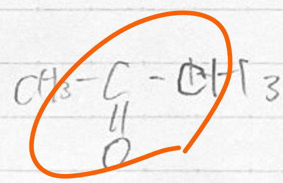
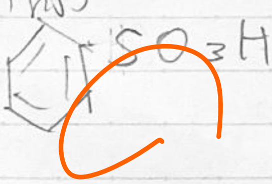


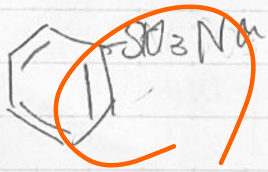
F


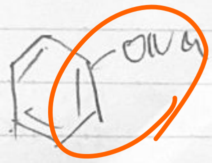
K


M


問3

N


O


P


4-17

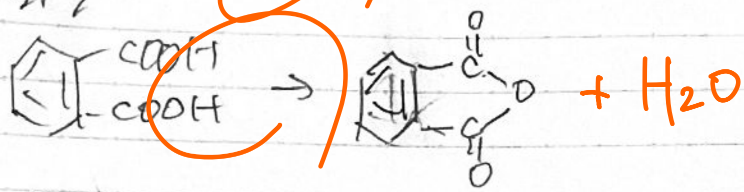
問1

あ C_5H_6O

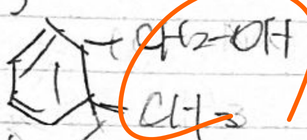
い 164

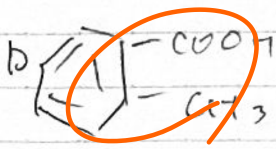
う $C_{10}H_{12}O_2$

問2

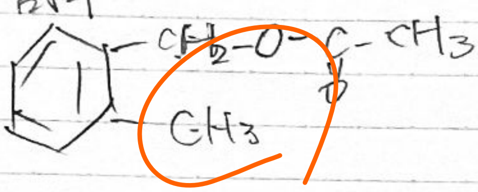


問3

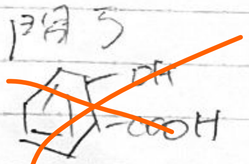
C 

D 

問4

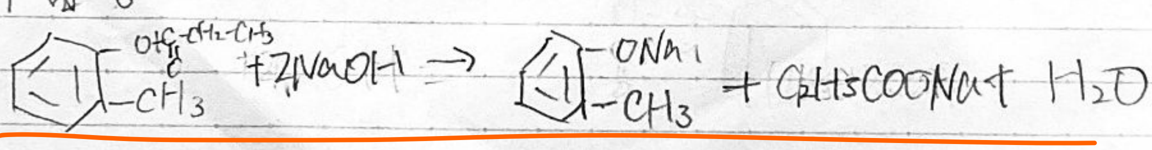


問5



パラ位のやつです

問6



パラ位になおす

2-回

問1

A C=C-C-C
1-ブテン

B C-C=C-C
2-ブテン

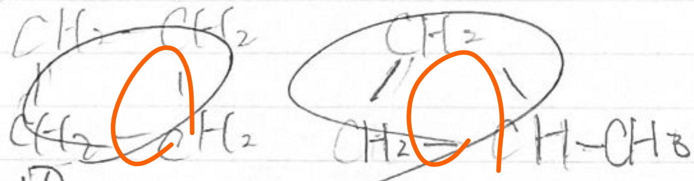
C C-C-C-C
ブタン

問2

D 2種類

E 1種類 3種類

問3



3-1回

問1

$C_{12}H_{20}O_6$

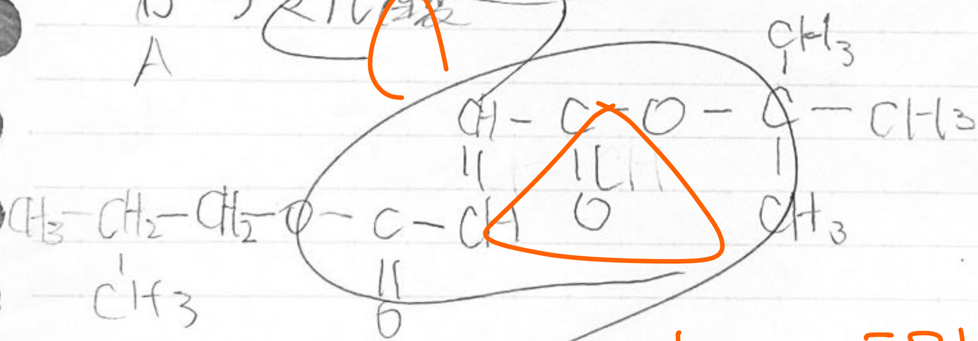
問2

B 2-メチル-1-プロパノール

C 2-メチル-2-プロパノール

D フマル酸

A

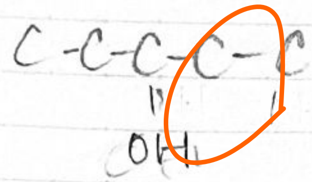


シス・トランス区別して

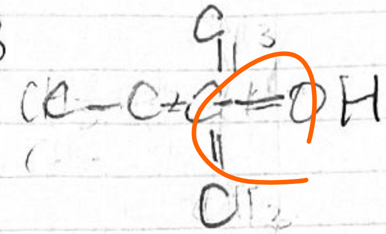
答えはトランス型

3-$\boxed{4}$

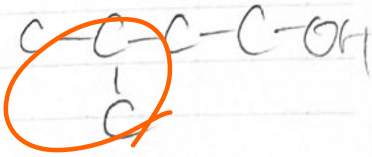
(1) A



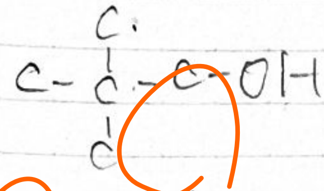
B



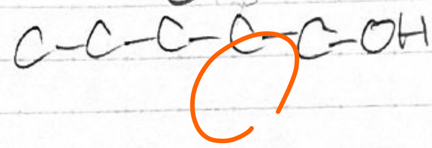
C



D

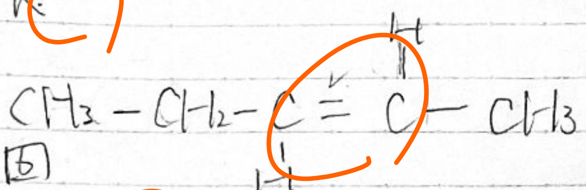


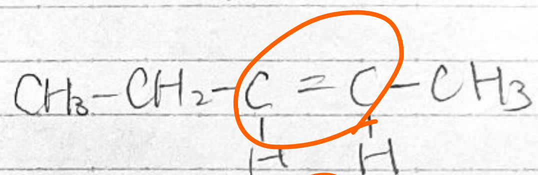
E



(2) K

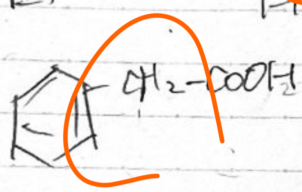
(3)



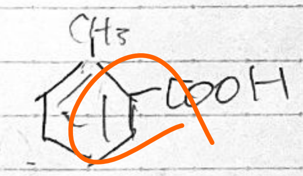


4-$\boxed{5}$

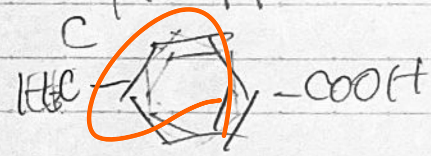
A



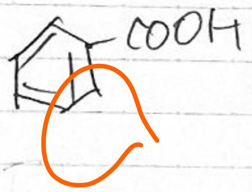
B



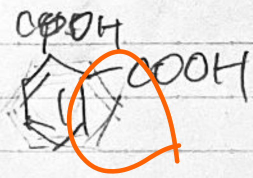
C



D



E



F